Fiscal Year:	FY 2025	**Task Last Updated:**	FY 06/04/2025
PI Name:	Lawler, John Ph.D.		
Project Title:	Upstream Regulation of Nox2 and Skeletal Muscle Atrophy During Microgravity and Countermeasure Development		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
PI Email:	jml2621@tamu.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	979-862-2038
Organization Name:	Texas A&M University		
PI Address 1:	Health & Kinesiology		
PI Address 2:	305 Gilchrist Bldg. 2929 Research Blvd. Redox Biology & Cell Signaling Laboratory		
PI Web Page:			
City:	College Station	**State:**	TX
Zip Code:	77843	**Congressional District:**	17
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2016-17 Space Biology (ROSBio) NNH16ZTT001N-FG. App G: Flight and Ground Space Biology Research
Start Date:	06/01/2019	**End Date:**	05/31/2026
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	3	**No. of Master' Degrees:**	1
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	8
No. of Bachelor's Candidates:	11	**Monitoring Center:**	NASA ARC
Contact Monitor:	Jones, Harry	**Contact Phone:**	650.604.5518
Contact Email:	harry.jones@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 05/31/2026 per the PI (Ed., 6/5/25). NOTE: End date changed to 05/31/2025 per NSSC information (Ed., 6/21/24). NOTE: End date changed to 05/31/2024 per NSSC information (Ed., 11/14/23). NOTE: End date changed to 05/31/2023 per F. Hernandez/ARC (Ed., 6/20/22). NOTE: Start/end dates changes to 6/1/2019-7/31/2022 (originally 3/1/2019-5/31/2022) per NSSC award documents per F. Hernandez/ARC (Ed., 12/3/2020)		
Key Personnel Changes/Previous PI:	August 2020 report: Dr. Khaled Kamal - Assistant Research Scientist ; Amin Mohajeri - PhD student; Mariam Othman - PhD student (add) Adding Joo Kim (PhD student) Jordyn Johnson - MS student. (remove - left because of COVID) Add: Aggie Research Scholars: Devon Roeming, Samhitha Harvey, Danielle DeCastro, Gracie Barrow, Yasmin Bagheri Victoria Frank, Anish Kumaran, Mickael Gergheis, Kyle Hawley, Fernanda Sifuentes (Aggie Research Scholar) 2022- Graduated: Myles McFarland (Aggie Research Scholar) 2020- ; Mollie Linder (Aggie Research Scholar) 2020- ; Hallie Harris (Aggie Research Scholar) 2020 - ; Francisco Melesio (Aggie Research Scholar) 2020 - ; Sonny Rodriguez (Aggie Research Scholar) 2020- ; Mia Ngyuen (Aggie Research Scholar) 2020- ; Lorrie Hill 2018-2020 (Kinesiology Research Scholar). ADD: Nina Kim, PhD student		

COI Name (Institution):	Fluckey, James Ph.D. (Texas A & M, College Station)
Grant/Contract No.:	80NSSC19K0432
Performance Goal No.:	
Performance Goal Text:	
Task Description:	Skeletal muscles are dynamic mechanical and metabolic machines that drive body movement and energy expenditure. Skeletal muscles are dynamic tissues that can adapt quickly to alterations in mechanical unloading by altering their mass and muscle fiber cross-sectional area. Skeletal muscle strength and endurance are essential to the health, well-being, and performance of astronauts during spaceflight and upon return to a gravitational environment. The mechanical unloading due to the microgravity (µG) of spaceflight causes muscle fiber atrophy and fiber-type shift of postural muscles in the lower extremities and flexors in the upper extremities. Microgravity also increases the risk of skeletal muscle damage, weakness, and thus the risk of injury upon reloading (e.g., extravehicular activities, Mars). Mechanotransduction is the ability to sense and regulate adaptive responses to increased or decreased loading. New paradigms have emerged from ground analogs of µG that have contributed to a leap of our understanding of mechanotransduction and muscle atrophy. Specifically, the mu-splice variant of neuronal nitric oxide synthase (nNOSµ) was discovered as causal in both muscle hypertrophy with overloading and atrophy with unloading. Our laboratory has found that reactive oxygen species (ROS) directly contribute to both muscle fiber atrophy and fiber-type shift from slow to fast. Pilot Data and cutting-edge research have identified mitochondria, the Nox2 isoform of NADPH oxidase, and upstream angiotensin II receptor 1 (AT1R) as sources of ROS during mechanical unloading. Preliminary Data show that inhibition of Nox2 translocation of nNOSµ away from the sarcolemma, muscle fiber atrophy, and fiber-type shift. However, the upstream mechanisms that regulate Nox2 during µG are poorly understood, impeding progress in space biology and novel countermeasure development. The lack of such knowledge impedes our development in understanding the mechanisms that underlie redox regulation of mechanotransduction in skeletal muscle. This grant application serves as a renewal and extension and Renewal of our research team's NNX13AE45G award, particularly stretching our horizons in understanding how Nox2 assembly is enhanced during microgravity in skeletal muscle. New studies have identified novel inhibitors for proteins recently govern assembly of the Nox2 complex at the cell membrane acid—sphingomyelinase (ASMase) and cyclophilin A. We hypothesize that the following novel countermeasures will protect against nNOSµ translocation and the spaceflight phenotype— (a) the ASMase inhibitor etidronate (Didronel) and (b) cyclophilin A inhibitor TMN-355. We further postulate that Nox2 is causal in ROS-induced suppression of anabolic signaling. The efficacy and specificity of the above countermeasures will be confirmed with gene knockdown experiments. Texas A&M is a rich research environment for NASA research, including the Space Life Science Program. We will use the latest molecular and image analysis tools in the development of highly novel countermeasures against spaceflight sarcopenia during microgravity. Dr. Lawler and Dr. Fluckey's laboratories have continued to be supported by NASA, and are dedicated to finding targeted, antioxidant countermeasures against spaceflight sarcopenia. The ground hindlimb unloading model will be used in short and long-term experiments. Our research will also directly translate to skeletal muscle wasting in clinical setting on Earth, an important mission of the Space Biology program. For example, hospitalization, particularly in an ICU (intensive care unit) can reduce skeletal muscle by 25%. Cast immobilization can decrease affected muscle mass by 30% as well. In addition, mechanical unloading due to disuse (e.g., bedrest) and illness (e.g., sepsis, chronic obstructive pulmonary disease, chronic heart failure) exacerbates atrophy, weakness, and impedes recovery.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our research will also directly translate to skeletal muscle wasting in clinical setting on Earth, an important mission of the Space Biology program. For example, hospitalization, particularly in an ICU (intensive care unit) can reduce skeletal muscle mass by 25%. Cast immobilization can decrease affected muscle mass by 30% as well. In addition, mechanical unloading due to disuse (e.g., bedrest) and illness (e.g., sepsis, chronic obstructive pulmonary disease, chronic heart failure) exacerbates atrophy, weakness, and impedes recovery.
Task Progress:	- Pectin and fish oil reduce damage, oxidative stress, and fibrosis in heart irradiated with heavy ion radiation experienced in space. - Micronized, biocompatible timed release ion matrix mitigates oxidative stress, damage, and disrruption of nitric oxide synthase in unloaded skeletal muscle. - A novel bioreactor for skeletal muscle hypertrophy and atrophy by manipulating uniaxial cyclic strain: proof of concept. Physiological alterations in anabolic and catabolic signaling reflect changes found in vivo. - Metformin-sensitive effects of a high fat diet on skeletal muscle morphology and sarcolemmal protein signaling in young mice. Nutrient dense diet promotes muscle hypertrophy. - NADPH oxidase-2 and RANKL form a positive feedback loop that drive pathology with Duchenne muscular dystrophy. - Unacylated ghrelin mitigates skeletal muscle damage and oxidative Stress in a Duchenne muscular dystrophy model.
Bibliography Type:	Description: (Last Updated: 11/13/2025)
Abstracts for Journals and Proceedings	Kendra J, Golpasandi S, Othman M, Kim, JH, Kamal KY, Lawler JM. "Micronized biocompatible ceramic promotes muscle IL-6 release in disuse." 2024 Texas American College of Sports Medicine (TACSM) Regional Meeting, Waco, TX, February 22-23, 2024. Abstracts. 2024 Texas American College of Sports Medicine (TACSM) Regional Meeting, Waco, TX, February 22-23, 2024. , Feb-2024

Abstracts for Journals and Proceedings	Gergeis M, Gabriel A, Othman, MO, Kim JH, Rauth R, Kama KY, Lawler JM. "RANKL inhibition mitigates Nox2 elevation, damage, and inflammation in dystrophic gastrocnemius muscle." Student Research Week, Texas A&M University, College Station, Texas, March 18-22, 2024. Abstracts. Student Research Week, Texas A&M University, College Station, Texas, March 18-22, 2024. , Mar-2024
Abstracts for Journals and Proceedings	Lawler JM, Kim JH, Othman MA, Kamal KY, Fluckey JD. "The sirtuin-1 agonist SRT2104 mitigates redox signaling and nNOS translocation in unloaded skeletal muscle." 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. Abstracts. 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. , Jan-2025
Abstracts for Journals and Proceedings	Lawler, JM. "New insights into mechanotransduction in Space City: Saying no to the school of hard Nox." Texas A&M Institute of Musculoskeletal Science Research Summit, College Station, TX, May 14-15, 2025. Abstracts. Texas A&M Institute of Musculoskeletal Science Research Summit, College Station, TX, May 14-15, 2025. , May-2025
Articles in Peer-reviewed Journals	Lawler JM, Kamal KY, Botchlett RE, Woo SL, Li H, Hord JM, Fluckey JD, Wu C. "Metformin ablates high fat diet-induced skeletal muscle hypertrophy and elevation of sarcolemmal GLUT4 when feeding is initiated in young adult male mice." Connect Tissue Res. 2025 Mar 7;1-15. https://doi.org/10.1080/03008207.2025.2471853 ; PMID: 40052722 , Mar-2025
Articles in Peer-reviewed Journals	Othman M, Kim J, Kamal K, Lawler J. "RANKL elevation contributes to skeletal muscle myopathy in aging: Unveiling of novel mechanisms." Physiology. 2025 May 12;40(S1):0567. https://journals.physiology.org/doi/abs/10.1152/physiol.2025.40.S1.0567 , May-2025
Articles in Peer-reviewed Journals	Lawler J, Kim J, Othman M, Kamal K, Fluckey J. "The sirtuin-1 agonist SRT2104 mitigates redox signaling and nNOS translocation in unloaded skeletal muscle." Physiology. 2025 May 12;40(S1):1343. https://journals.physiology.org/doi/abs/10.1152/physiol.2025.40.S1.1343 , May-2025
Journal/Magazine covers	Lawler JM, Kamal KY, Botchlett RE, Woo SL, Li H, Hord JM, Fluckey JD, Wu C. "Cover in the journal Connective Tissue Research for article, 'Metformin ablates high fat diet-induced skeletal muscle hypertrophy and elevation of sarcolemmal GLUT4 when feeding is initiated in young adult male mice.'" Connect Tissue Res. 2025 Mar 7;1-15. https://doi.org/10.1080/03008207.2025.2471853 ; PMID: 40052722 , Mar-2025
Journal/Magazine covers	Mason CE, Green J, Adamopoulos KI, Afshin EE, Baechle JJ, Basner M, Bailey SM, Bielski L, Borg J, Borg J, Broddrick JT, Burke M, Caicedo A, Castañeda V, Chatterjee S, Chin C, Church G, Costes SV, De Vlaminck I, Desai RI, Dhir R, Diaz JE, Etlin SM, Feinstein Z, Furman D, Garcia-Medina JS, Garrett-Bakelman F, Giacomello S, Gupta A, Hassanin A, Houerbi N, Irby I, Javorsky E, Jirak P, Jones CW, Kamal KY, Kangas BD, Karouia F, Kim J, Kim JH, Kleinman A, Lam T, Lawler JM, Lee JA, Limoli CL, Lucaci A, MacKay M, McDonald JT, Melnick AM, Meydan C, Mieczkowski J, Muratani M, Najjar D, Othman MA, Overbey EG, Paar V, Park J, Paul AM, Perdyan A, Proszynski J, Reynolds RJ, Ronca AE, Rubins K, Ryon KA, Sanders LM, Glowe PS, Shevde Y, Schmidt MA, Scott RT, Shirah B, Sienkiewicz K, Sierra M, Siew K, Theriot CA, Tierney BT, Venkateswaran K, Hirschberg JW, Walsh SB, Walter C, Winer DA, Yu M, Zea L, Mateus J, Beheshti A. "Cover in the journal Nature for article, 'A second space age spanning omics, platforms, and medicine across orbits.'" Nature. 2024 Aug;632(8027):995-1008. https://doi.org/10.1038/s41586-024-07586-8 ; PMID: 38862027 , Aug-2024
Papers from Meeting Proceedings	Kendra J, Golpasandi S, Naman A, Othman M, Kim J, Rauth R, Moustafa K, Lawler J, Morton A. "Micronized biocompatible ceramic promotes muscle derived IL-6 release in disuse." 2024 Texas American College of Sports Medicine (TACSM) Regional Meeting, Waco, TX, February 22-23, 2024. International Journal of Exercise Science: Conference Proceedings. 2024;2(16):99. https://digitalcommons.wku.edu/ijesab/vol2/iss16/99 , Feb-2024